



TECHNICAL DESIGN DOCUMENT

TravelPort

Architecture, Data Design & Integration Guide

VERSION

v1.0.0

BACKEND

.NET 8 Clean Architecture

FRONTEND

React 18 + TypeScript

DATABASE

SQL Server + EF Core 8

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7.1 Razorpay

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7.2 SMTP Email

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7.3 Duffel (Optional)

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Clean Architecture Overview

Layer responsibilities and dependency rules

1.1 Layer Responsibilities

API (Presentation) — Controllers, Middleware (JWT, exception handling, logging), Action Filters, Program.cs (DI root)

depends on ↓

Application (Business Logic) — AuthService, FlightService, HotelService, BookingService, WalletService, AdminService; DTOs, FluentValidation validators, ICacheService, IUnitOfWork

depends on ↓

Domain (Core) — Entities (User, Flight, Hotel, Booking, Wallet, WalletTransaction, Coupon, RefreshToken), Enums (BookingStatus, BookingType, TravelClass), ValueObjects — ZERO external dependencies

Infrastructure & Persistence implement interfaces defined in ↑

Persistence — TravelPortDbContext, Repositories (Flight, Hotel, Booking, Wallet, Coupon, User, Admin), EF Core Migrations, DataSeeder

Infrastructure — JwtService, CacheService, SmtpeMailService, RazorpayService, DuffelProvider; BackgroundServices (BookingExpiryWorker, RefreshTokenCleanupWorker)

1.2 Dependency Rule

```
Domain ← Application ← Infrastructure ← API
      ↑           ↑
      Persistence External APIs
```

```
# Arrows point INWARD. Outer layers know about inner layers.
# Inner layers (Domain) know nothing about outer layers.
# This is enforced via project references in .csproj files.
```

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Backend Design

Entities, services, repositories and caching

2.1 Core Domain Entities

ENTITY	KEY FIELDS	NOTES
User	<code>Id, Email, PasswordHash, Role, IsBlocked, WalletId</code>	Wallet created on registration via cascade
Flight	<code>Id, FlightNumber, Origin, Destination, Airline, DepartureTime, ArrivalTime, AvailableSeats, BasePrice</code>	900+ seeded across 42 routes
Hotel	<code>Id, Name, City, StarRating, PricePerNight, Amenities, RoomTypes</code>	60+ across 12 cities
Booking	<code>Id, BookingReference (FL/HT prefix), UserId, FlightId?, HotelId?, FinalAmount, Status, BookingType</code>	Status: Pending → Confirmed / Cancelled
Wallet	<code>Id, UserId, Balance</code>	One wallet per user; balance validated before deduction
WalletTransaction	<code>Id, WalletId, Amount, Type (Credit/Debit), Description, ReferenceId</code>	Immutable audit trail
Coupon	<code>Id, Code, DiscountType (Percentage/Fixed), DiscountValue, MinOrderValue, IsActive, CouponType</code>	Flight-specific or Hotel-specific
RefreshToken	<code>Id, UserId, Token, ExpiresAt, IsRevoked</code>	Cleaned up by background worker every 24h

2.2 Caching Strategy

CACHE KEY PATTERN	TTL	INVALIDATED ON
<code>flights:{Origin}:{Destination}:{Date}:{Pax}:{Class}</code>	5 min	Flight booking (seat count change)
<code>flight:{id}</code>	30 min	Flight booking

CACHE KEY PATTERN	TTL	INVALIDATED ON
<code>hotels:{City}:{In}:{Out}:{Guests}:{Rating}:{Sort}</code>	5 min	Hotel booking
<code>hotel:{id}</code>	30 min	Hotel booking

`ICacheService` wraps `IDistributedCache` with JSON serialization. Default: `AddDistributedMemoryCache()`. Production swap: replace with `AddStackExchangeRedisCache()` in Program.cs — zero service-layer changes.

2.3 Atomic Booking + Wallet

```
// FlightService.BookAsync - simplified flow
1. ValidateFlightExists(flightId)
2. CheckSeatsAvailable(flight, passengerCount)
3. couponDiscount = ApplyCoupon(couponCode, baseAmount) // optional
4. if (useWallet) _wallet.DeductAsync(userId, finalAmount) // NO SaveChanges yet
5. booking = new Booking { ... Status = Pending }
6. _bookings.AddAsync(booking)
7. flight.AvailableSeats -= passengerCount
8. _unitOfWork.SaveChangesAsync() // ← ONE atomic commit: booking + wallet + seats
9. _cache.RemoveAsync("flights:{key}")
10. _email.SendBookingConfirmed(booking)
```

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Frontend Architecture

Feature modules, state management and API integration

3.1 Feature Module Structure

```
frontend/src/
├── api/
│   ├── axios.ts           ← Axios instance + JWT interceptor (auto-refresh)
│   └── endpoints.ts       ← All API URL constants (no hardcoded URLs elsewhere)
├── components/
│   ├── common/            ← Button, Select, Modal, Skeleton, ConfirmDialog
│   └── ui/                ← Design system primitives
├── features/
│   ├── auth/              ← authSlice (Redux), LoginForm, RegisterForm
│   ├── flights/           ← FilterSidebar, FlightCard, FarePopup, TravellerDetails
│   ├── hotels/            ← HotelCard, HotelFilters, RoomSelector
│   ├── bookings/         ← BookingCard, BookingDetailPage, PDFDownload
│   └── admin/             ← AdminPage (4 tabs: Dashboard, Users, Bookings, Coupons)
├── pages/                 ← Route-level components (FlightsPage, HotelsPage, etc.)
├── services/              ← flightService, hotelService, bookingService, userService
├── store/                 ← Redux store + authSlice
├── types/index.ts         ← All shared TypeScript interfaces
└── routes/                ← AppRouter, PrivateRoute, AdminRoute
```

3.2 JWT Interceptor Pattern

```
// api/axios.ts
api.interceptors.response.use(
  (res) => res,
  async (error) => {
    if (error.response?.status === 401 && !originalRequest._retry) {
      originalRequest._retry = true;
      const token = await refreshAccessToken(); // POST /auth/refresh
      api.defaults.headers.common['Authorization'] = `Bearer ${token}`;
      return api(originalRequest);
    }
    return Promise.reject(error);
  }
);
```

3.3 Key Design Choices

DECISION	CHOICE	REASON
State management	<code>Redux Toolkit</code>	Predictable auth state; DevTools support; minimal boilerplate
Validation	<code>Zod</code>	Runtime type safety matching backend FluentValidation contracts
Styling	<code>Tailwind CSS v3</code>	Utility-first; no CSS-in-JS overhead; purged in production
Build tool	<code>Vite 6</code>	Fast HMR; native ESM; proxy config for /api → backend
HTTP	<code>Axios</code>	Interceptor support for JWT refresh; typed response wrapper
Named exports	<code>All components</code>	Avoids default-export aliasing confusion at import sites

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Database Design

Core tables, relationships and migration strategy

4.1 Entity Relationships

RELATIONSHIP	CARDINALITY	FK
User → Wallet	1 : 1	Wallet.UserId (Cascade delete)
User → RefreshToken	1 : many	RefreshToken.UserId
User → Booking	1 : many	Booking.UserId
Wallet → WalletTransaction	1 : many	WalletTransaction.WalletId
Flight → Booking	1 : many (nullable)	Booking.FlightId
Hotel → Booking	1 : many (nullable)	Booking.HotelId
Hotel → RoomType	1 : many	RoomType.HotelId

4.2 Migration Strategy

EF Core Code-First migrations. Auto-applied on API startup via `db.Database.Migrate()` in `Program.cs`. DataSeeder runs after migration — idempotent checks prevent re-seeding on restart.

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Authentication Flow

JWT + Refresh token rotation

1

Login

POST /auth/login → validate credentials (BCrypt) → issue JWT (15 min) + RefreshToken (7 days, stored in DB)

2

API Call

Client sends Authorization: Bearer {access_token} header

Axios interceptor catches 401 → POST /auth/refresh → new access token issued

4

Token Rotation

Old refresh token revoked; new pair issued. IsRevoked flag set in DB.

5

Cleanup

RefreshTokenCleanupWorker deletes revoked/expired tokens every 24 hours

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External Integrations

RAZORPAY

Payment gateway. Enabled via config flag. Mock fallback in dev returns success.

SMTP / OFFICE365

Transactional email. HTML table layouts for Gmail/Outlook compatibility. TLS 587.

DUFFEL

Real flight search API. Disabled by default — rich DB seed data used instead.